RC Car Calibration



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Parameter Setting

Parameter Setting

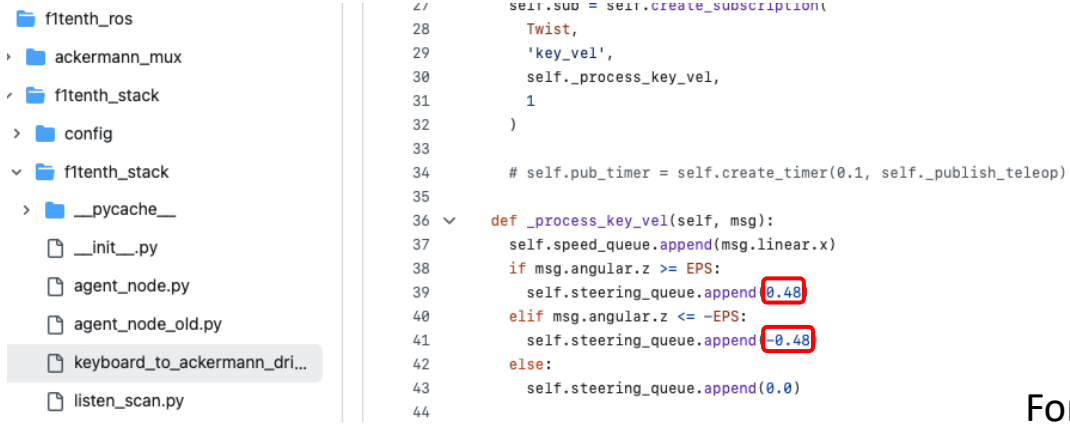
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1. Change forward/backward rate to **2.0**



It makes the default velocity to be 2m/s.

2. Change steering value to **0.6 / -0.6**



For max steering angle.

Parameter Setting

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3. Change '*steering_angle_to_servo_gain*' to **-1.0 or 1.0**
    - Please set the value to either -1.0 or 1.0, ensuring the steering direction matches the arrow keys when controlled via keyboard.



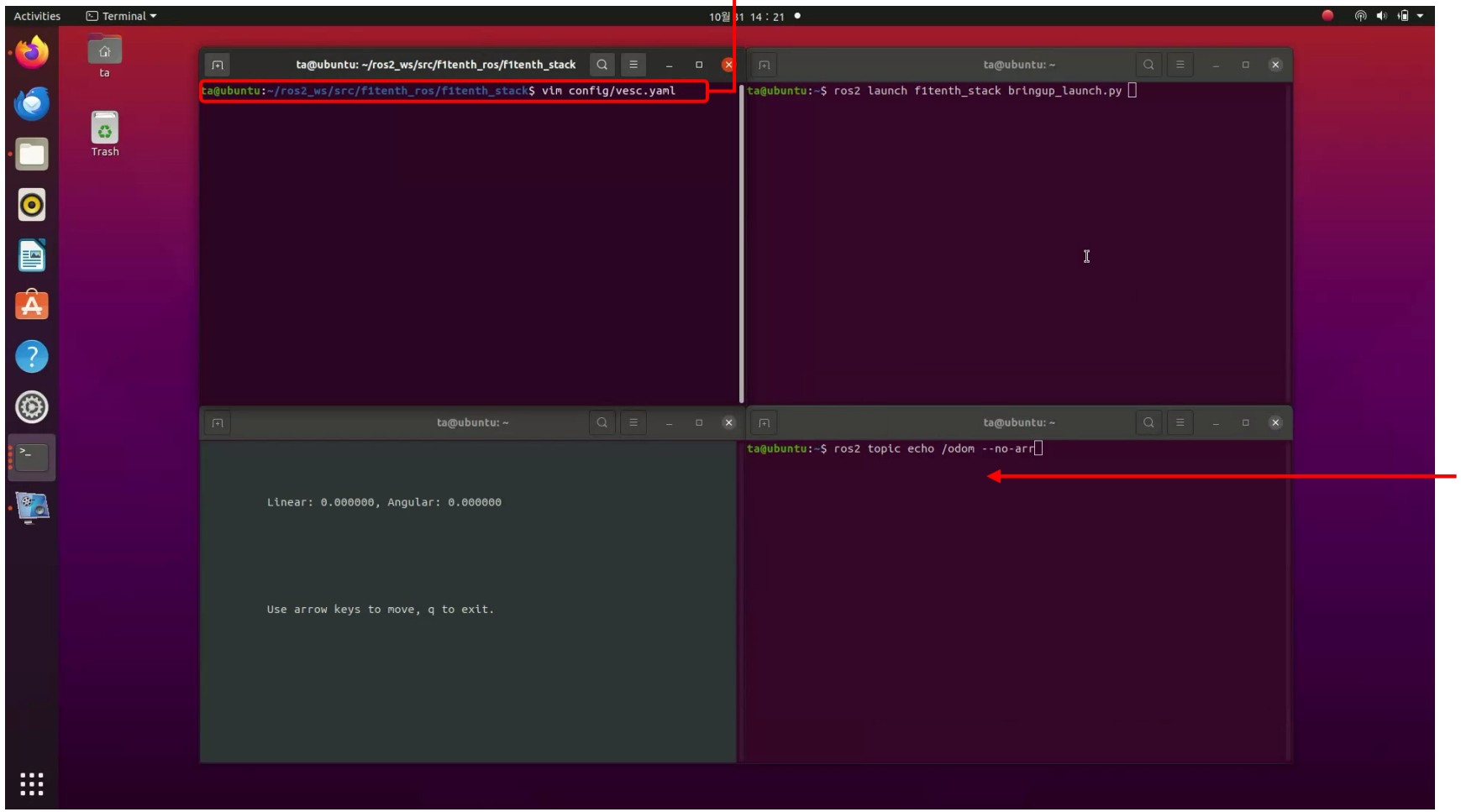
Odometry - Position

Odometry - Position

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- when RC-Car is **not** moving

in f1tenth_ros/f1tenth_stack/config/vesc.yaml
change ***'speed_to_erpm_offset'***

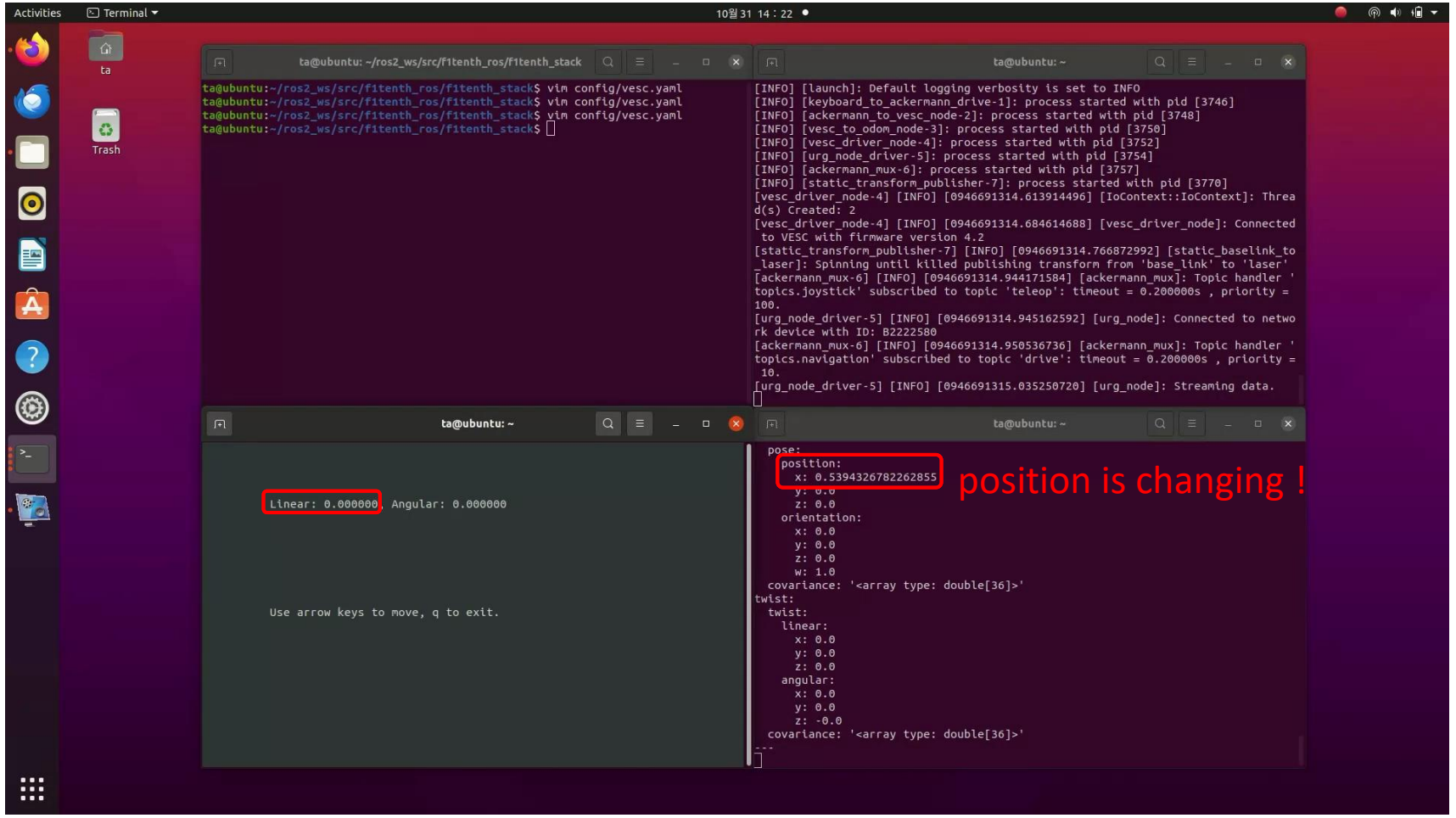


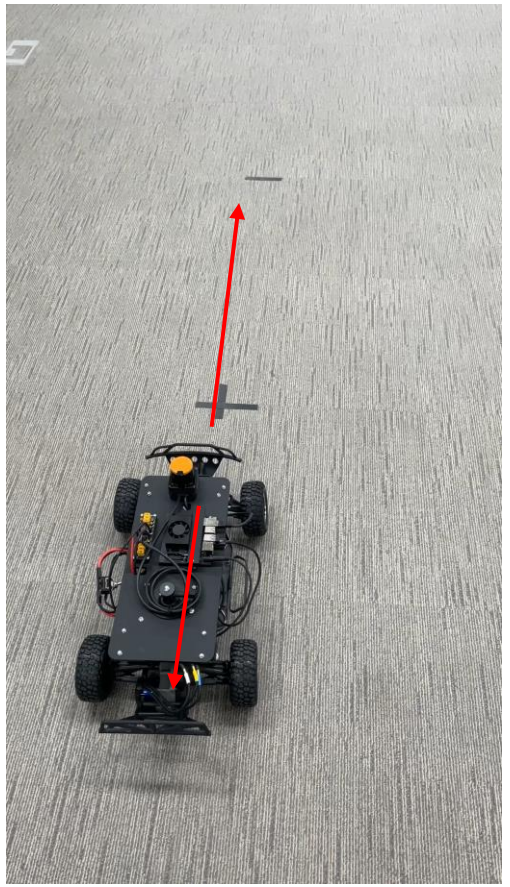
position is not changing !

Odometry - Position

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- when RC-Car is **moving**



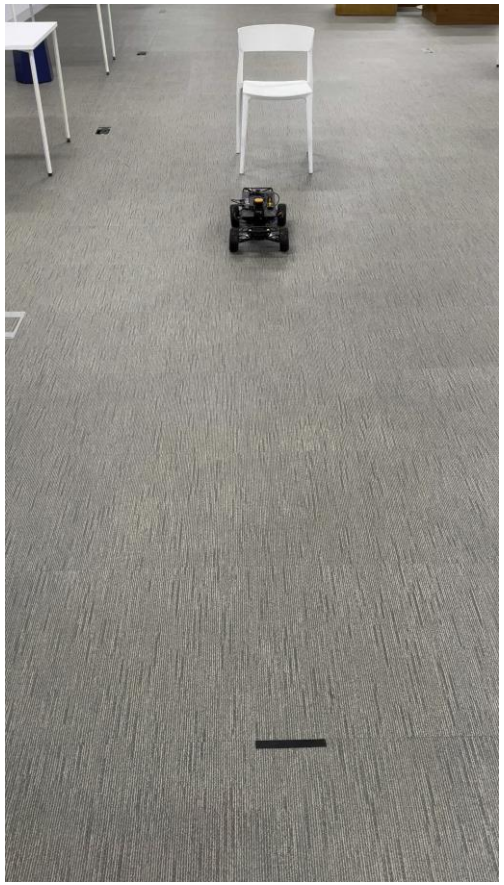


Odometry - Direction

Odometry - Direction

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(Fail) RC-Car go straight





When it goes **left/right**, **decrease/increase** the value.

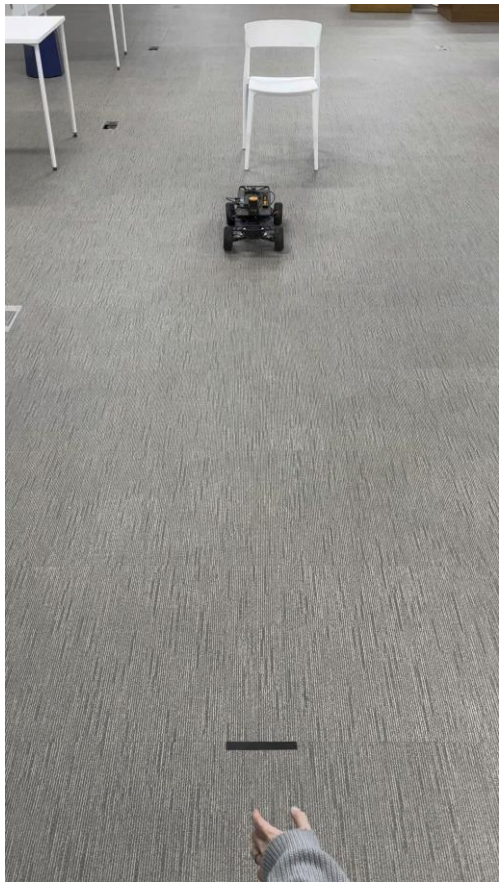
When it goes **left/right**, **increase/decrease** the value.

Adjust it according to your specific vehicle.

In this case, it goes **right**.

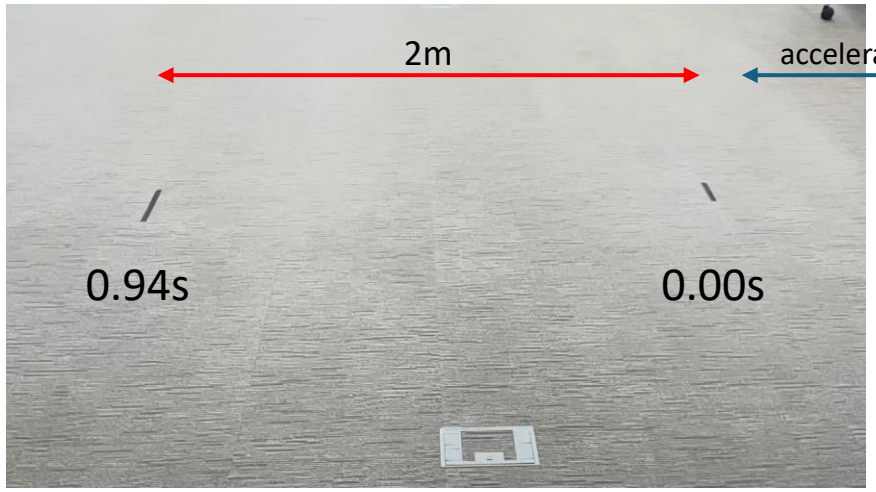
=> We **decrease** the value (0.494 -> 0.48)

(Success) RC-Car go straight



Control - Velocity

Control - Velocity



Little fast



Well fitted !



When it is **faster than 2m/s**, **decrease** the value.

When it is **slower than 2m/s**, **increase** the value.

In this case, it is **faster** than 2m/s.

=> We **decrease** the value (4000 -> 3750)

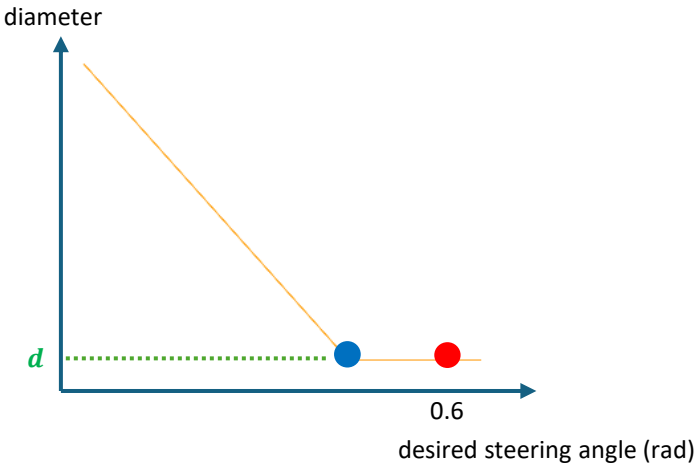
Control - Steering

Control - Steering (real)

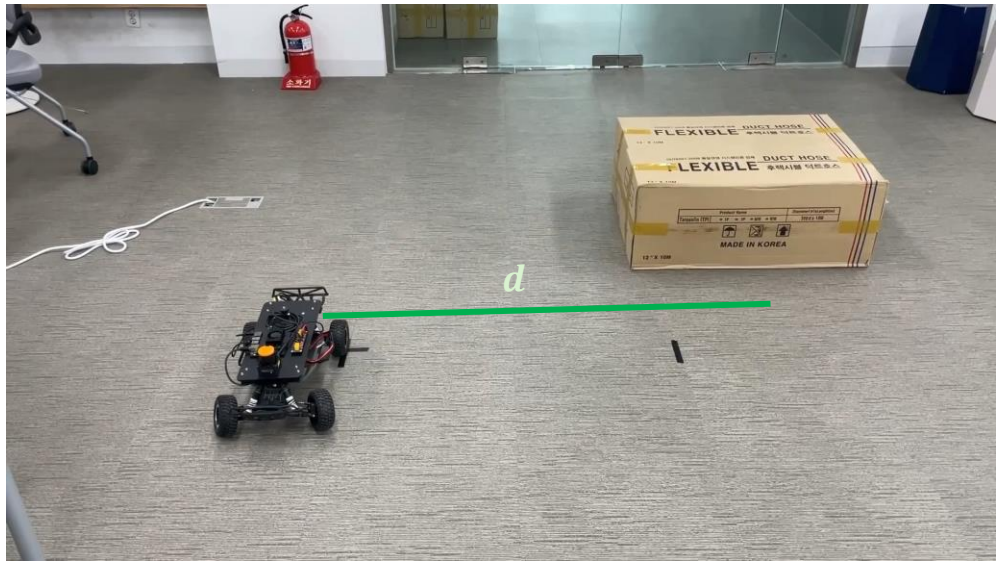
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We need to find the diameter at the blue point which means the maximum steering angle.

1. To do that, we find the minimum diameter.



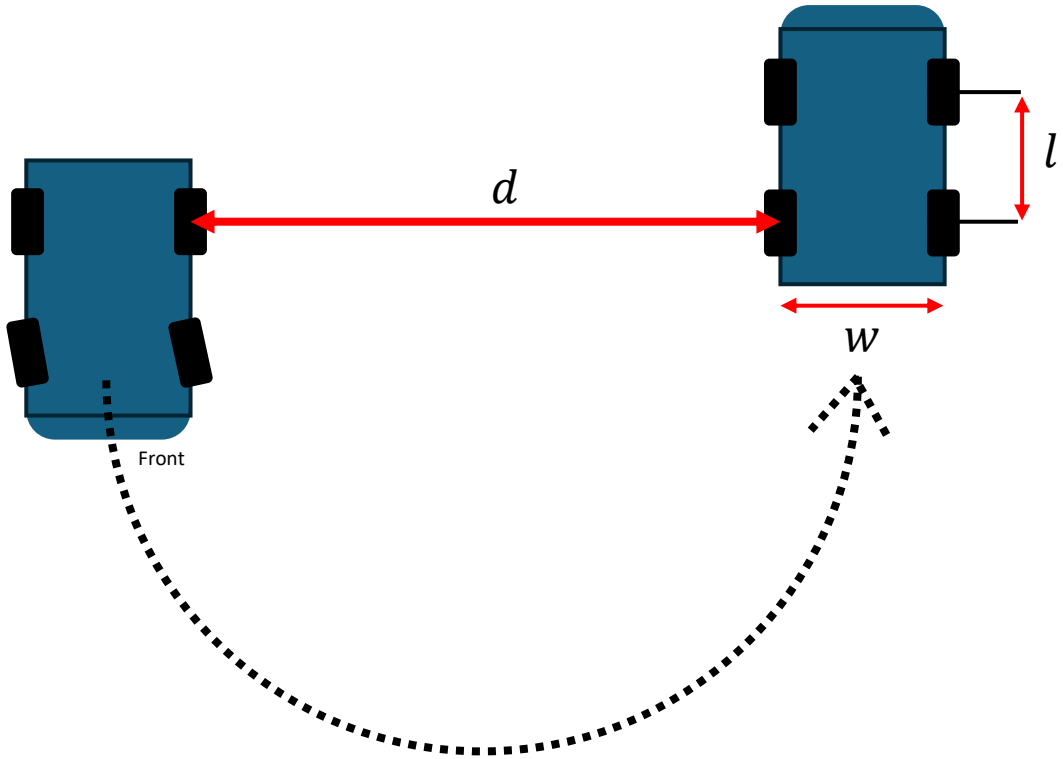




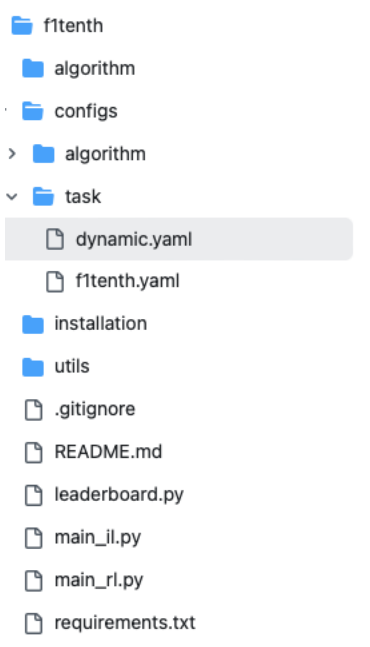
Control - Steering (real)

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2. After you get the diameter of the inner circle,
   you should calculate the maximum steering angle (δ).



$$\delta = \tan^{-1}\frac{2l}{d+w}$$




```yaml
# dynamic system
integrator: rk4
model: st

# dynmaic configs
mu: 1.0489
C_Sf: 4.718
C_Sr: 5.4562
length_f: 0.2
length_r: 0.15
h: 0.074
m: 3.74
I: 0.04712
s_min: -1.0
s_max: 1.0
sv_min: -2.65
sv_max: 2.65
v_switch: 7.319
a_max: 9.51
v_min: -8.0
v_max: 8.0
width: 0.29
length: 0.57
lidar_pos_offset: 0.155
```

$l = 0.2 + 0.15 = 0.35$

$w = 0.29$

Measure your own values or use pre-defined values.

Control - Steering (sim)

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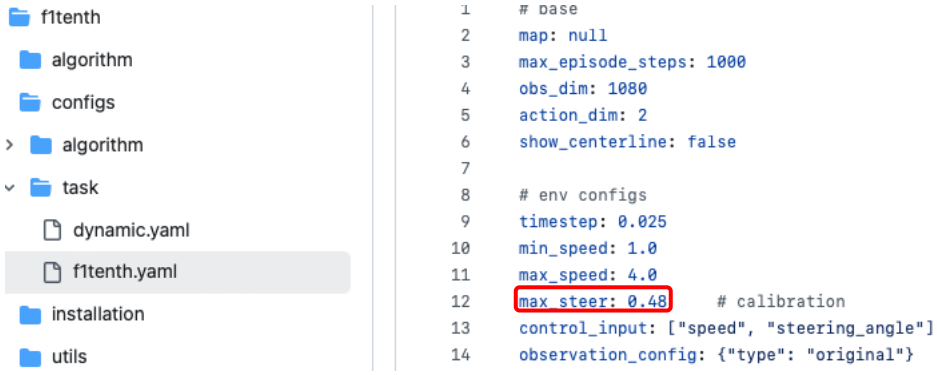
3. Then change the desired steering angle to your measurement (δ).





Same diameter !

4. If the diameter does not change too much, put the maximum steering angle to the environment.



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Thank you for listening !